Study Buddy

Learn more, study efficiently

Research and Demo

Introduction

Study Buddy is an Alexa app that allows students to automatically generate relevant questions from a lecture video. Alexa then quizzes them and explains the answer.

Goal and Problem Addressed

Goal: Help students learn the material taught on lectures better and aid them to prepare for exams efficiently

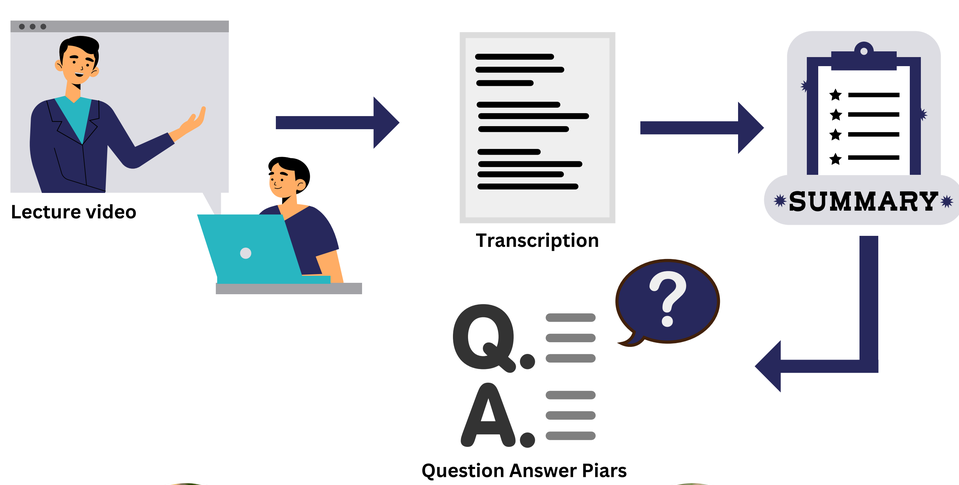
Problems: Students needs peer to ask questions, and they need to generate questions manually. Time consuming to rewatch lecture.

Audience

- College students
- K-12 Students
- Professors
- Parents

Alexa, quiz me on Lecture 4 from CIS 521

Pipeline to automatically generate QA pairs



Methods

Developing and applying cutting-edge NLP research

Summarization Model: Adrian Wang's Fine-tuned GPT-3 Model

Question Generation Model: Bootstrapped GPT-3 and fine-tuned model. Please look at our research paper on the QR code for details.

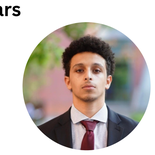
Alexa integration

1. Student prompts quiz for class A, lecture B
2. QA's are fetched from database
3. Student is asked a question
4. Student gives answer
5. Students asks for explanation

Differentiation

Before	After
Student needs to prepare questions after lecture.	Relevant questions are automatically generated
Student needs a peer to ask them questions	Student can study by themselves at any time
Student isn't provided with specific explanations	Student is provided with specific explanations

Hannah Gonzalez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